Spam Email Detection using Machine Learning

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1. Objective

The main objective of this project is to build a simple machine learning model that can automatically identify whether an email is spam or not spam.

2. Introduction

In today's digital world, email is one of the most widely used communication tools. However, many unwanted emails known as spam are received daily. These emails may contain advertisements, fake offers, or even harmful links. Artificial Intelligence (AI) and Machine Learning (ML) provide smart solutions to such problems. In this project, a basic ML model is used to detect spam emails based on their content.

3. Theory

Spam detection is a classification problem in machine learning where emails are divided into two categories: Spam (unwanted emails) and Ham (normal emails).

This project uses the Naive Bayes algorithm, which is based on probability. It is widely used for text classification because it is simple, fast, and effective.

Since machine learning models cannot understand text directly, the text is converted into numerical form using a technique called Count Vectorization.

4. Methodology

Step 1: Collect a small dataset of emails.
Step 2: Label the emails as spam (1) or not spam (0).
Step 3: Convert text data into numerical form.
Step 4: Train the machine learning model.
Step 5: Test the model with new emails.
Step 6: Analyze the prediction results.

5. Implementation

The project is implemented in Python using the Scikit-learn library. The Multinomial Naive Bayes algorithm is used for classification.

6. Observation

It is observed that spam emails often contain words like "win", "offer", and "earn". The model uses these patterns to make predictions. However, with a small dataset, the accuracy may not be very high.

7. Result

The model is able to classify emails as spam or not spam. Although the results are not perfect due to limited data, the model demonstrates how machine learning works.

8. Advantages

- Easy to implement
- Fast processing
- Useful for real-world applications

9. Limitations

- Requires large dataset for better accuracy
- May give incorrect predictions
- Assumes all features are independent

10. Conclusion

This project shows how machine learning can be used to solve real-life problems like spam detection. With better data and improvements, the model can become more accurate and reliable.

11. Applications

- Email filtering systems
- SMS spam detection
- Fraud detection
- Cybersecurity